

Literature Review: EEG-Based Classification of Dementia

Project focus: Alzheimer's disease (AD), frontotemporal dementia (FTD), and cognitively normal controls (CN).

Purpose and background

This review summarizes research that motivates the project and explains the main methodological choices. Resting-state electroencephalography (EEG) records electrical activity from electrodes on the scalp. Researchers study its frequency content, temporal complexity, and relationships between electrodes to look for differences among dementia groups. These measurements are relatively accessible, but an EEG pattern is not a diagnosis on its own. The central research question here is whether a model can distinguish AD, FTD, and CN in people it has never seen before.

The source dataset

Miltiadous and colleagues published the open dataset used in this project [1]. It contains eyes-closed resting EEG from 88 people: 36 with AD, 23 with FTD, and 29 CN. The recordings use 19 scalp electrodes at 500 Hz. The release includes both raw EEG and preprocessed derivatives. The latter have already undergone filtering, rereferencing, Artifact Subspace Reconstruction (ASR), independent component analysis (ICA), and removal of components identified as likely artifacts. Starting from those derivatives avoids treating raw acquisition noise as if it had not been handled, while still leaving residual quality questions for a project-specific audit. The small number of FTD participants is an important constraint: many windows can be cut from their recordings, but those windows do not create more independent patients.

What EEG features have shown

Rostamikia and colleagues extracted time-domain, frequency-domain, complexity, and connectivity features from the same 88-person dataset [2]. They found potentially discriminative electrode and frequency-band patterns; their support vector machine reported 87.8% accuracy for AD versus FTD and 93.5% for dementia versus controls under tenfold cross-validation. Those are two binary tasks, whereas our main task has three classes. Their findings support testing ordinary spectral features as a reference and examining connectivity, but the reported accuracies cannot be used as a direct benchmark for a different task and validation design.

Wang and colleagues also analyzed these 88 participants and separated periodic oscillations from the aperiodic background of the power spectrum [3]. They reported that aperiodic parameters added useful information for AD-versus-FTD discrimination. This matters because a compact representation built only from conventional band power can miss useful structure. Their analysis does not establish that every additional feature will generalize in a small cohort; each candidate must be tested with training-only feature selection and held-out participants.

Mlinaric and colleagues studied functional connectivity features with leave-one-subject-out validation [4]. Their AD-versus-FTD distinction was harder than separating either dementia group from controls, with a reported stacked-model ROC-AUC of 65.10% for AD versus FTD. ROC-AUC is not the same metric as our three-class balanced accuracy. The paper reinforces the need to report results by diagnosis and to resist comparing headline percentages from different tasks as though they measured the same thing.

Why tensor decomposition is relevant

EEG naturally has several axes, such as participant, electrode, frequency, and time. Flattening these into one long feature vector can make their relationships less explicit. Gupta and colleagues used canonical polyadic tensor decomposition to find a compact set of interpretable time-space-frequency EEG patterns [5]. Their study concerned stages of cognitive impairment, including mild cognitive impairment and Alzheimer's dementia, rather than the three groups in our dataset. It provides a reason to test tensor factors, not evidence that a particular factorization must outperform simpler features here. A decomposition can preserve common signal variation while still losing subtle clues that separate AD from FTD.

Validation is central

Miltiadous and colleagues compared epoch-based tenfold cross-validation with leave-one-patient-out evaluation for dementia EEG classification [6]. Scores dropped substantially when entire people were excluded from training. Their explanation is that epochs from one person resemble one another, so an epoch-level split can reward recognition of an individual rather than recognition of a disease pattern. The implication for this project is direct: all windows from one person must remain together, and preprocessing fits, feature selection, decomposition, and model tuning must use training participants only.

Synthesis and project gap

Existing work suggests that EEG contains useful information, but performance depends strongly on the diagnostic task, features, and validation unit. This project therefore compares simple spectral descriptions with tensor, waveform, and connectivity representations under participant-level evaluation, while reporting balanced accuracy and recall for each diagnosis. The current evidence does not justify clinical use or an accuracy ceiling. A future external cohort would be necessary to test whether a selected approach works beyond people whose data have already been explored.

References

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